

A1	2020	☒	☐
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The function $f(x) = x^3$ is transformed to $g(x) = (x-2)^3 + 5$ by a horizontal translation of 2 units followed by a vertical translation of 5 units.

Which row of the table shows the directions of the translations?

	Horizontal translation of 2 units	Vertical translation of 5 units
A.	Left	Up
B.	Right	Up
C.	Left	Down
D.	Right	Down

A2	2020	☒	☐
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The circle $x^2 - 6x + y^2 + 4y - 3 = 0$ is reflected in the x -axis.

Sketch the reflected circle, showing the coordinates of the centre and the radius.

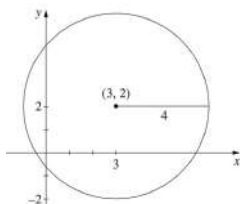
$$x^2 - 6x + y^2 + 4y - 3 = 0$$

$$x^2 - 6x + 9 + y^2 + 4y + 4 = 3 + 9 + 4$$

$$(x-3)^2 + (y+2)^2 = 16$$

The original circle has centre $(3, -2)$ and radius 4.

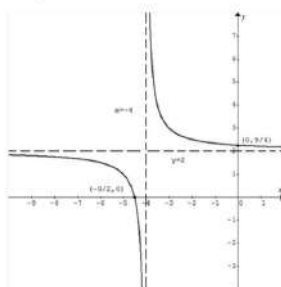
The reflected circle has centre $(3, 2)$ and radius 4.



A3	2021	☒	☐
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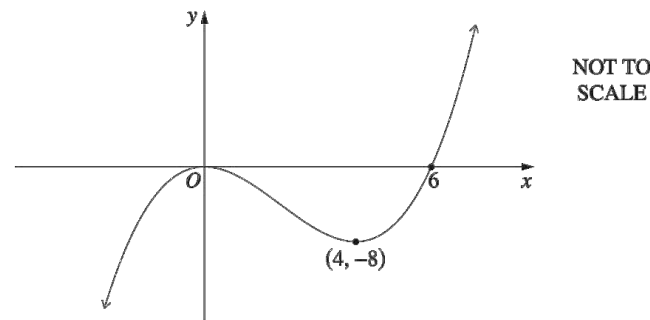
Without using calculus, sketch the graph of $y = 2 + \frac{1}{x+4}$, showing the asymptotes and the x and y intercepts.

Translate $y = \frac{1}{x}$ left by 4 units and up by 2 units.



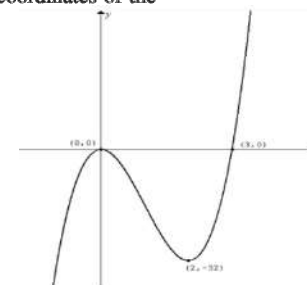
A4	2021	☒	☐
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Consider the graph of $y = f(x)$ as shown.



Sketch the graph of $y = 4f(2x)$ showing the x -intercepts and the coordinates of the turning points.

Q21 Dilate $y = f(x)$ vertically by a factor of 4 and horizontally by a factor of $\frac{1}{2}$. $(0, 0)$ remains the same, $(0, 6)$ becomes $(0, 3)$, and $(4, -8)$ becomes $(2, -32)$.



A5	2022	☒	☐
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The graph of the function $f(x) = x^2$ is translated m units to the right, dilated vertically by a scale factor of k and then translated 5 units down. The equation of the transformed function is $g(x) = 3x^2 - 12x + 7$.

Find the values of m and k .

$$g(x) = k(x-m)^2 - 5$$

$$= k(x^2 - 2xm + m^2) - 5$$

$$= kx^2 - 2mkx + m^2k - 5$$

By inspection $k = 3$

By equating coefficient of x

$$2mk = 12$$

$$6m = 12$$

$$m = 2$$

ALTERNATE SOLUTION

$$g(x) = k(x-m)^2 - 5$$

$$g(x) = 3x^2 - 12x + 7$$

$$= 3(x^2 - 4x) + 7$$

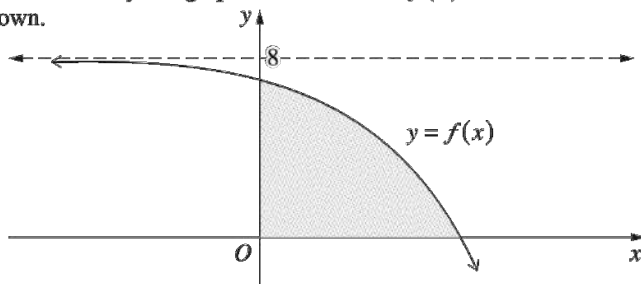
$$= 3(x^2 - 4x + 4) + 7 - 12$$

$$= 3(x-2)^2 - 5$$

So $k = 3$ and $m = 2$

A6	2021	☒	☐
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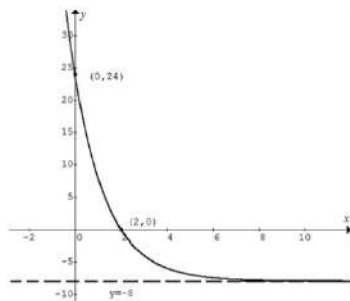
The region bounded by the graph of the function $f(x) = 8 - 2^x$ and the coordinate axes is shown.



A new function $g(x)$ is found by taking the graph of $y = -f(-x)$ and translating it by 5 units to the right.

Sketch the graph of $y = g(x)$ showing the x -intercept and the asymptote.

Q28b $g(x)$ is the reflection of $f(x)$ in both axes and then translation by 5 units to the right.



A7	2022	☒	☐
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Let $f(x) = \sin(2x)$.

Find the value of x , for $0 < x < \pi$, for which $f'(x) = -\sqrt{3}$ AND $f''(x) = 2$.

Q25 $f(x) = \sin 2x$, $f'(x) = 2 \cos 2x = -\sqrt{3}$ AND $f''(x) = -4 \sin 2x = 2$

$\therefore \cos 2x = -\frac{\sqrt{3}}{2}$ AND $\sin 2x = -\frac{1}{2}$, $0 < x < \pi$

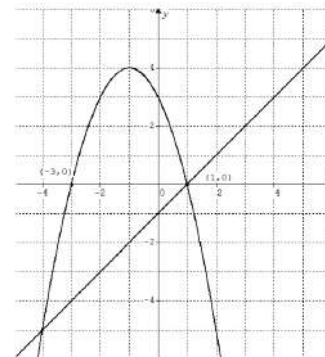
For $\cos 2x = -\frac{\sqrt{3}}{2}$, $x = \frac{5\pi}{12}, \frac{7\pi}{12}$. For $\sin 2x = -\frac{1}{2}$, $x = \frac{7\pi}{12}, \frac{11\pi}{12}$

Only $x = \frac{7\pi}{12}$ satisfies both equations.

A8	2023	☒	☐
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(a) Sketch the graphs of the functions $f(x) = x - 1$ and $g(x) = (1 - x)(3 + x)$ showing the x -intercepts.

(b) Hence, or otherwise, solve the inequality $x - 1 < (1 - x)(3 + x)$.

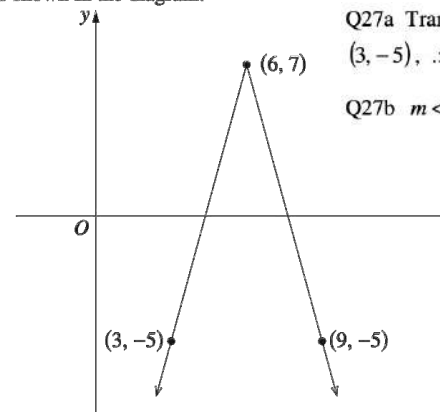


Q19b At the intersections, $x - 1 = (1 - x)(3 + x)$, $x = -4$ or $x = 1$
 $x - 1 < (1 - x)(3 + x)$ for $-4 < x < 1$

Q20 $\theta - 60^\circ = -60^\circ, 240^\circ, 300^\circ$
 $\therefore \theta = 0^\circ, 300^\circ, 360^\circ$

A9	2023	☒	☐
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The graph of $y = f(x)$, where $f(x) = a|x - b| + c$, passes through the points $(3, -5)$, $(6, 7)$ and $(9, -5)$ as shown in the diagram.



Q27a Translations: $b = 6$, $c = 7 \therefore y = a|x - 6| + 7$

$(3, -5)$, $\therefore a = -4$

Q27b $m < \frac{7}{6}$ and $m > \frac{7 - (-5)}{6 - 9} = -4 \therefore -4 < m < \frac{7}{6}$

(a) Find the values of a , b and c .

(b) The line $y = mx$ cuts the graph of $y = f(x)$ in two distinct places.

Find all possible values of m .

A10	2024	☒	☐
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The parabola $y = (x - 3)^2 - 2$ is reflected about the y -axis. This is then reflected about the x -axis.

What is the equation of the resulting parabola?

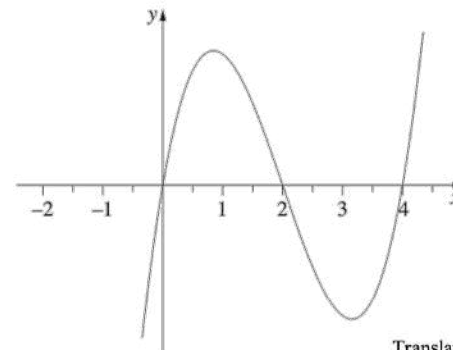
- A. $y = (x + 3)^2 + 2$ Q4 $y = (x - 3)^2 - 2 \rightarrow y = (-x - 3)^2 - 2 \rightarrow -y = (-x - 3)^2 - 2$
 B. $y = (x - 3)^2 + 2$ $\therefore y = -(x + 3)^2 + 2$
 C. $y = -(x + 3)^2 + 2$
 D. $y = -(x - 3)^2 + 2$

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A11	2024	☒	☐
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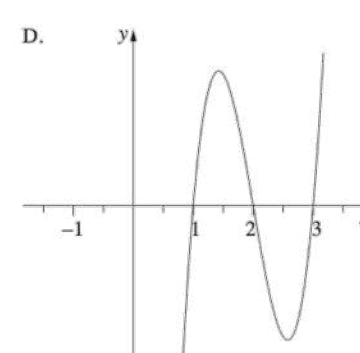
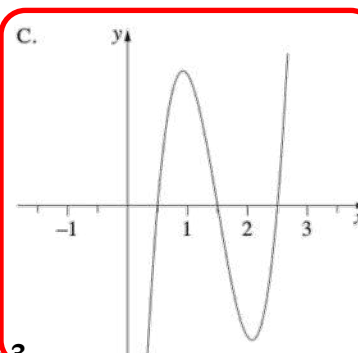
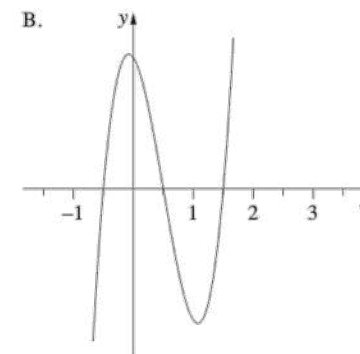
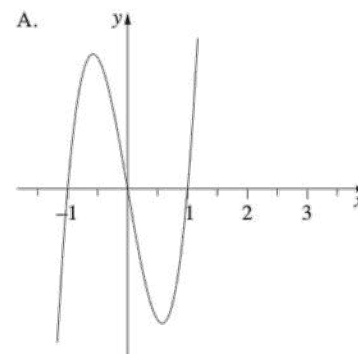
The diagram shows the graph $y = f(x)$.

1



Translate to the right by 1 and then dilate by $\frac{1}{2}$ horizontally.

Which of the following best represents the graph $y = f(2x - 1)$?





B1	2020	☒	☐
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Which interval gives the range of the function $y = 5 + 2\cos 3x$?

A. $[2, 8]$

B. $[3, 7]$

C. $[4, 6]$

D. $[5, 9]$

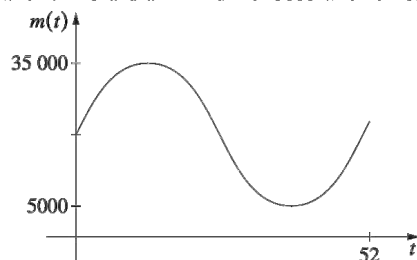
$$[5 - 2, 5 + 2] = [3, 7]$$

B2	2022	☒	☐
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The population of mice on an isolated island can be modelled by the function

$$m(t) = a \sin\left(\frac{\pi}{26}t\right) + b,$$

where t is the time in weeks and $0 \leq t \leq 52$. The population of mice reaches a maximum of 35 000 when $t = 13$ and a minimum of 5000 when $t = 39$. The graph of $m(t)$ is shown.



(a) What are the values of a and b ?

(b) On the same island, the population of cats can be modelled by the function

$$c(t) = -80 \cos\left(\frac{\pi}{26}(t - 10)\right) + 120.$$

Consider the graph of $m(t)$ and the graph of $c(t)$.

Find the values of t , $0 \leq t \leq 52$, for which both populations are increasing.

(c) Find the rate of change of the mice population when the cat population reaches a maximum.

$$\text{Q31a } b = \frac{35000 + 5000}{2} = 20000, \quad a = 35000 - 20000 = 15000$$

$$\text{Q31b } m(t) \text{ is increasing for } 0 < t < 13, \quad c(t) \text{ is increasing for } 10 < t < 36. \text{ Both are increasing for } 10 < t < 13.$$

$$\text{Q31c } c(t) \text{ reaches maximum at } t = 36 \text{ and}$$

$$\frac{dm}{dt} = 15000 \left(\frac{\pi}{26}\right) \cos\left(\frac{\pi}{26} \times 36\right) \approx -643, \text{ i.e. decreasing by 643 per week approximately.}$$

B3	2021	☒	☐
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For what values of x , in the interval $0 \leq x \leq \frac{\pi}{4}$, does the line $y = 1$ intersect the graph of $y = 2\sin 4x$?

$$\text{Q20 Let } 2\sin 4x = 1 \text{ and } 0 \leq x \leq \frac{\pi}{4} \text{ i.e. } 0 \leq 4x \leq \pi$$

$$\therefore \sin 4x = \frac{1}{2} \text{ and } 4x = \frac{\pi}{6}, \frac{5\pi}{6} \therefore x = \frac{\pi}{24}, \frac{5\pi}{24}$$

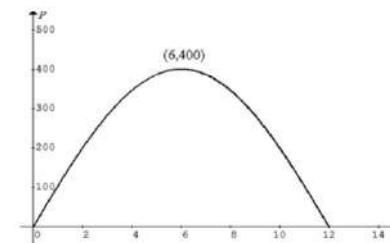
B4	2021	☒	☐
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Kenzo has a solar powered phone charger. Its power, P , can be modelled by the function

$$P(t) = 400 \sin\left(\frac{\pi}{12}t\right), \quad 0 \leq t \leq 12,$$

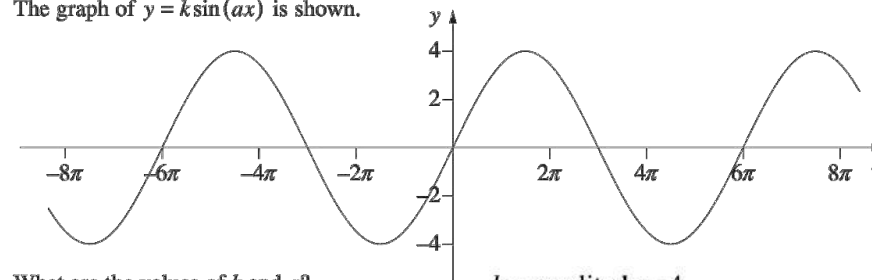
where t is the number of hours after sunrise.

Sketch the graph of P for $0 \leq t \leq 12$.



B5	2022	☒	☐
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The graph of $y = k \sin(ax)$ is shown.



What are the values of k and a ?

$$k = \text{amplitude} = 4$$

$$\text{Period} = \frac{2\pi}{a}$$

$$6\pi = \frac{2\pi}{a}$$

$$\therefore a = \frac{1}{3}$$

B8

2024

✓

✗

Anna is sitting in a carriage of a Ferris wheel which is revolving. The height, $A(t)$, in metres above the ground of the top of her carriage is given by

$$A(t) = c - k \cos\left(\frac{\pi t}{24}\right),$$

where t is the time in seconds after Anna's carriage first reaches the bottom of its revolution and c and k are constants.

2

1

3

Q28a Amplitude $= k = \frac{39-3}{2} = 18$
 $A = 3$ when $t = 0 \therefore c - k = 3 \therefore c = 21$

Q28b Period $T = \frac{2\pi}{\frac{\pi}{24}} = 48$

Q28c Same heights $B(t) = A(t) \therefore \cos\left(\frac{\pi}{24}(t-6)\right) = \cos\left(\frac{\pi}{24}t\right)$
 $\frac{\pi}{24}(t-6) = \pm \frac{\pi}{24}t$, choose $\frac{\pi}{24}(t-6) = -\frac{\pi}{24}t$, $t = 3$
 $\therefore B(3) = A(3) = 21 - 18 \cos\frac{\pi}{8} \approx 4.37$
 $\frac{1}{2}T$ later, $t = 3 + 24 = 27$, $B(27) = A(27) = 21 - 18 \cos\frac{7\pi}{8} \approx 37.63$

The top of each carriage reaches a greatest height of 39 metres and a smallest height of 3 metres.

(a) Find the value of c and k . 2

(b) How many seconds does it take for one complete revolution of the Ferris wheel? 1

(c) Billie is in another carriage. The height, $B(t)$, in metres above the ground of the top of her carriage is given by 4

$$B(t) = c - k \cos\left(\frac{\pi}{24}(t-6)\right),$$

where c and k are as found in part (a).

During each revolution, there are two occasions when Anna's and Billie's carriages are at the same heights. At what two heights does this occur? Give your answer correct to 2 decimal places.